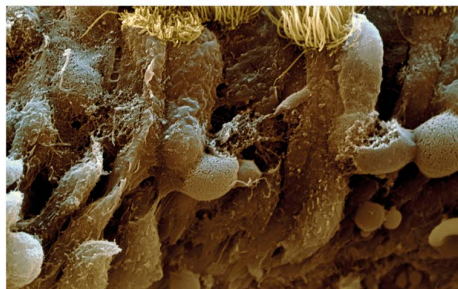


SMELL, TASTE, AND TOUCH

RECEPTORS THAT SENSE PRESSURE, PAIN, AND TEMPERATURE ARE WIDESPREAD IN THE BODY. TASTE AND SMELL, IN CONTRAST, ARE “SPECIAL SENSES” BECAUSE THEIR RECEPTORS ARE COMPLEX AND LOCALIZED, AND DETECT SPECIFIC STIMULI.

SMELL

Smell is a sense that can detect chemical molecules known as odorants floating in the air. Specialized epithelial tissue provides a smelling zone, known as the olfactory epithelium, on the roof of the nasal cavity. In humans, smell is more sensitive than taste and may be able to distinguish millions of odours. Smell is important for warning of dangers, such as smoke and poisonous gas, and for appreciating food and drink. The sense of smell tends to deteriorate with age, so young people are able to distinguish a wider range of odours and experience them more vividly than older people.

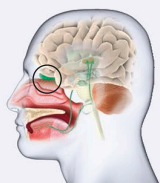


NASAL LINING

Epithelial cells in the lining of the nasal chamber have tufts of hair-like cilia, which wave germ- and odorant-trapping mucus towards the back of the chamber to be swallowed.

HOW WE SMELL

Odour molecules dissolve in the mucus lining the nasal chamber. In the roof of the chamber, they touch the cilia (microscopic, hair-like endings of olfactory receptor cells). If the correct molecule slots into the same-shaped receptor on the cilia membrane, like a key in a lock, a nerve impulse is generated. The impulses are partly processed by intermediate neurons called glomeruli in the olfactory bulb.



LOCATION

